

claude-windows / 8d4866bf-8edd-4da9-909e-022ddee12675

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\8d4866bf-8edd-4da9-909e-022ddee12675\subagents\workflows\wf_4e32c8b8-725\agent-a61e84dff401a6f3e.jsonl`  
- SHA-256: `6a789893720225f9f606007bcce5edcb5a1588e3bb7754a891c7b562fe8f8500`  
- Source modified: `2026-06-19T17:19:52+00:00`  
- Imported at: `2026-07-05T16:48:25+00:00`  
- project: `wf_4e32c8b8-725`  
- session_id: `8d4866bf-8edd-4da9-909e-022ddee12675`
```

Transcript

1. user (2026-06-19T17:18:10.921Z)

You are auditing the docs/ folder of the badminton-highlight-indexer repo to decide which docs can move to docs/archives/. Be rigorous and EVIDENCE-BASED ? the owner has run two doc-staleness sweeps and cares deeply about not archiving live work or breaking links.

REPO ARCHIVE POLICY (docs/archives/README.md) ? APPLY STRICTLY:

- Archive ONLY work in a clear TERMINAL state: DONE / COMPLETED / REJECTED / fully SUPERSEDED. "Never archive in-flight or pending work."
- KEEP-LIVE (do NOT archive) anything that is: living/active, in-progress, a not-yet-started but still-intended design/plan (owner-gated future work is LIVE, not terminal), a reference/navigation doc (CODE_MAP, README, HOW_RALLY_DETECTION_WORKS, EVALUATION), a template (PROJECT_DOC_TEMPLATE), or a roadmap/backlog/strategy doc still being tracked.
- Destinations inside archives/: past_projects/ (bounded completed project / tech-debt / starter effort) · research/ (completed or superseded research notes/plans) · decisions/ (ADRs) · checkpoints/ (dated state snapshots) · archives/ top-level (one-off audits/reports/launch records). Naming convention for a completed doc: <NAME>-COMPLETED-<date>.md.
- ALREADY-ARCHIVED COPIES EXIST for these (so the top-level file is likely a pointer-stub or a stale DUPLICATE to reduce, NOT a fresh move): docs/archives/CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md, docs/archives/DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md, docs/archives/HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md. If your doc already has an archived copy, check whether the top-level is a stub (leave it) or a full duplicate (flag DEDUPE).
- CONTRADICTION TRAP: a doc may say "COMPLETED" yet still be cited as a LIVE source elsewhere (e.g. CLAUDE.md cites DEFERRED_HARDENING_PLAN.md #16 as the next platform slice). Before recommending ARCHIVE, grep CLAUDE.md, docs/NEXT_STEPS.md, docs/README.md, and other top-level docs for the filename. If it is referenced as a live/next source, verdict = ASK_OWNER (note the contradiction), not ARCHIVE_NOW.

YOUR BATCH (read each from docs/<name>): PLATFORM_ARCHITECTURE.md, HOSTING_PLAN.md, UI_PROPOSAL.md, UI_ALPHA_EXECUTION_PLAN.md, STORAGE_SHARING_MODEL.md, VIDEO_LOCALITY_MODEL.md, COMMERCIALIZATION.md

For EACH doc:

1. Read the status header + TL;DR/intro + (if present) the \$7/\$8 progress tracker + definition-of-done. You do NOT need to read every line of long docs ? sample enough to judge lifecycle state honestly.
2. Decide lifecycle and verdict per the strict policy. Remember: owner-gated / not-yet-started DESIGN docs are LIVE future work (KEEP_LIVE), NOT terminal. Only DONE/COMPLETED/REJECTED/fully-superseded is terminal.
3. Check for an already-archived copy (the policy lists three; also Glob docs/archives for the doc's stem). If the top-level is a thin pointer-stub ? KEEP_LIVE (it's intentional). If it's a FULL DUPLICATE of an archived COMPLETED copy ? DEDUPE_STUB (recommend reducing top-level to a stub or removing).
4. Cross-check "cited as live": Grep CLAUDE.md, docs/NEXT_STEPS.md, docs/README.md, and the other top-level docs/ for this filename. If it's cited as a live/next/authoritative source, it

CANNOT be archived even if it says COMPLETED ? ASK_OWNER with the contradiction noted.

5. Inbound references: Grep the whole repo (docs/, backend/, CLAUDE.md, *.py) for the filename to count link-breakage risk; list up to 6 key referrers.

Return the structured object with one entry per doc in your batch. Be decisive: most reference/template/active/design docs are KEEP_LIVE; reserve ARCHIVE_NOW for genuinely terminal single-efforts/reports/completed projects, and ASK_OWNER for the contradictory/ambiguous ones.

2. assistant (2026-06-19T17:18:18.827Z)

I'll start by reading the archive policy and the key cross-reference files, then audit each doc in my batch.

3. user (2026-06-19T17:18:22.181Z)

Your configuration specifies to merge with the ref 'refs/heads/docs/court-floor-rally-end' from the remote, but no such ref was fetched.

---ARCHIVES---

CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md

COMMERCIAL_READINESS_REVIEW.md

DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md

HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md

MONETIZATION_AUDIT.md

README.md

TEST_RUN_SUMMARY.md

checkpoints

decisions

past_projects

quality-iterations

research

4. user (2026-06-19T17:18:22.221Z)

```
1      # archives/
2
3      Umbrella directory for all historical and completed material.
4
5      ## Live vs Historical
6      - Live docs (for ramp-up, constant updates, and current work) live at the top level
of `docs/`.
7      - Everything here is no longer the primary source of truth. It is kept for context,
decision history, and regression detection.
8
9      ## Classification inside archives/
10     - `past_projects/` ? Completed small tech debt, starter projects, or low-regret
initiatives (e.g. the original low-regret moves + scaffolding work). These are "past" once they
reach a terminal state.
11     - `decisions/` ? Historical ADRs and major architectural decisions.
12     - `research/` ? Completed or superseded research notes and plans.
13     - `checkpoints/` ? Snapshots of major project states (especially useful for detecting
regressions in quality, priorities, or baselines over time as we do quality iterations).
14     - Other one-off audits, etc.
15
16     ## Archival Rule
17     Only move something into `archives/` (or its subdirectories) when the work is in a clear
terminal state:
18     - DONE
19     - COMPLETED
20     - REJECTED
21
22     Never archive in-flight or pending work.
23
24     When archiving a project/phase:
25     1. Ensure the main document has an explicit STATUS: COMPLETED (or DONE/REJECTED)
```

header at the top.

```
26      2. Link back to the current living documents in `../` (e.g. `COMMERCIALIZATION.md`,
`BACKLOG.md`).
27      3. For regression detection value, consider adding a dated entry under `checkpoints/`.
28
29      See `past_projects/README.md` for the specific guidance on that category.
30
```

5. assistant (2026-06-19T17:18:24.270Z)

Let me read the key cross-reference files and start reading my batch docs.

6. user (2026-06-19T17:18:25.225Z)

```
1      # Next steps (live) ? owned-model + multisport
2
3      **Updated:** 2026-06-13 (**rally-detection quality track**: fusion + experiment-harness
design docs merged,
4      owner-gated ? see the dedicated section below, the desktop-agent pick-up). Prior:
2026-06-10 (audit?M0?Gemini-battery session; later same day the **UI-alpha track opened**:
5      ADR-009 ratified ? KhelSutra Guru / khelsutra.guru, private `khelsutra` repo, local-
first delivery ? see
6      `docs/UI_ALPHA_EXECUTION_PLAN.md`; **PR-1 async jobs (#16) MERGED 2026-06-11, PR #87**;
**PR-2
7      upload/download MERGED 2026-06-11, PR #99** ? live-E2E-verified (3-part chunked upload
MD5 byte-identity +
8      compile?browser-download); next: PR-3 identity. Same day: the **8?9 quality sweep
#90?#98 landed** ? ruff/mypy/
9      coverage-floor CI lanes, SQLite conn-close fix, config unknown-key warning, hybrid-twin
collapse into
10     `trajectory_hybrid.py`, WSL tilde+abspath live-path fixes; suite 388 green; LOVO 0.626
re-verified exact).
11     **Authoritative roadmap:**
12     `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` (+ ADR-008 for topology). **Latest blow-by-
blow:** memory
13     `current-state.md` / `session-handoff.md`. This file = the prioritized "what to do
next," refreshed each session.
14
15     ## ? Next-session / GPU-box pickup (2026-06-15)
16     **M0 in-container quick wins are COMPLETE:** ship-gate provisional blocker (**#170**),
train/eval split guard (**#171**), the **doc-consistency sweep** (**#174** ? corrected 16
drifted docs vs the code), and the **P3 person-feature fusion litmus machinery is already
present + suite-green** (`backend/eval/fusion_compare.py` + `ablation.py`: person-driven ?F1 /
bootstrap CI / paired sign test; the real-golden `skipif` litmus waits only on GPU-extracted
data). ? CI is red repo-wide due to a **GitHub Actions minutes/billing outage** (account-level,
not code) ? work is local-gated + `--admin`-merged until Actions is restored (Settings ? Billing
? Actions minutes / spending limit).
17
18     **Immediate next step ? the real P3 LOVO measurement** (this box has a GPU +
`torch+cu124` + `cv2`; videos and trajectory CSVs are present, but the GPU-extracted golden
features + `rallies.csv` labels are not yet here):
19     1. **[owner / GPU box]** `python -m backend.eval.fusion_golden --manifest
output/human_lovo_manifest.json --out-dir "%USERPROFILE%\OneDrive\rally-annotator\golden-
data\features\2026-06-15-n6" ? produces `*_golden_features.json` for the 6 golden videos. (GPU
+ WSL + videos + trajectory CSVs + `rallies.csv` labels are owner-side only.)
20     2. Artifacts land in the **shared cloud folder** `C:\Users\avidu\OneDrive\rally-
annotator\golden-data\` (OneDrive auto-syncs both ways). Convention + workflow:
**[`GOLDEN_DATA_SHARING.md`](GOLDEN_DATA_SHARING.md)**; tracked in **#175**.
21     3. **[CPU dev/agent session]** `python -m backend.eval.fusion_compare --features-dir
<shared>\features\2026-06-15-n6 --record` + `python -m backend.eval.ablation --features-dir
<shared>\... --granularity group` ? real person-feature ?F1; copy the JSONs into `output/` to
also pass the `skipif` litmus tests (`tests/test_ablation.py::test_litmus_reproduces_gate_a`).
22
23     **Also queued (owner-gated / pending input):**
24     - Ship-gate target calibration ? **#172** (owner decision; needs corpus growth).
```

25 - Tests reorganization ? blocked on the owner's grouping metric (provide it ? an agent will do it).

26 - Restore CI ? the Actions outage is account-level (billing/minutes), not a repo/config bug.

27

28 ****Leaving (bulky / M2 ? no start without owner go):**** ActionFormer (M0.4), event-index DB migration (M0.1), Gemini-silver purge (M0.3), cost-to-Gen-2 benchmark + ByteTrack deprecation, multisport, repo rename, PMF execution.

29

30 **## Where we are (one paragraph)**

31 ADR-008 ratified ("repo split driven by productionization, not isolation"): training lives in-repo (`training/`)

32 behind a CI-enforced import wall; the featurizer is single-sourced (`backend/features/rally\_seq.py`, train==serve);

33 the ****Gen-0 harness ships**** (`python -m training.gen0.harness --manifest output/human\_lovo\_manifest.json --floor`)

34 eval_baselines/heuristic_lovo_n6.json --version <semver>) and produced artifact v0.1.0 (LOVO ****0.626****, floor passed,

35 sign test 5/6 wins p=0.109 ? honestly not significant, ship=False). The ****n=6** owned corpus is in the collector**

36 (262 rallies, Proprietary, commercial export = owned data only after the ShuttleSet reclassification). The ****Gemini**

37 battery is done** (see table) ? the moat is quantified.

38

39 **## The quality table that now drives strategy (IoU 0.5 vs human golden labels)**

40 | Path | Mahadevpura (clean-ish) | Kushagra (multi-court) |

41 |---|---|---|

42 | Heuristic (velocity windowing) | 0.657 | 0.157 |

43 | ****Gen-0 owned head**** (LOVO, n=6) | 0.744 | 0.561 |

44 | Two-stage WASB+Gemini refine | 0.83 | ? |

45 | ****Gemini wrapper**** (`gemini-flash-latest`) | ****0.93**** | ****0.66**** |

46

47 ****Reading:**** clean footage is commoditized (serve it with Gemini for cents). ****Multi-court** drops the commodity

48 wrapper to 0.66 ? that is the owned model's ship target** (its FPs are background-game pickups + dead-time merges,

49 the exact contrast-metric confound). Gen-0 is 0.10 behind with only n=6 ? a corpus-growth-sized gap, not an

50 architecture-sized one. Two-stage refine = the cost-efficient serving path for LONG tapes (uploads candidate clips

51 only); wrapper 503-flakiness (observed all session) makes the provider-fallback registry load-bearing.

52 Baselines tracked in `eval\_baselines/` (heuristic_lovo_n6.json, gemini_wrapper_2026-06-10.json).

53

54 **## Rally-detection quality track (design merged 2026-06-13, owner-gated; implementation in progress)**

55 ****This is the track a desktop/local agent is slated to pick up.**** Design docs landed (PR #112 + session). All implementation is ****owner-gated**** (per explicit in the plan); each step = ONE PR off `master`.

56

57 ****Hardening loop (T1-T5 positive tests + audit + finale) COMPLETE (2026-06-14, PRs #156/#157/#160/#161/#163 + #165 + final records).**** See archived

58 `docs/archives/CODE\_AUDIT\_AND\_TEST\_HARDENING-COMPLETED-2026-06-14.md` (Living Log with every pre/post/review) and `docs/archives/HARDENING\_LOOP\_HANDOFF-COMPLETED-2026-06-14.md` (full rules + STATE). Top-level now pointer stubs per #165 review. All 5 items + verification + archive done; clean slate.

59

59 - ****Docs:**** `docs/MULTI\_SIGNAL\_FUSION\_PLAN.md` (shuttle + person + audio, one fusion layer) .

60 `docs/EXPERIMENT\_HARNESS.md` (A/B + shadow + promotion gate, zero-regression) .

61 `docs/ROORA\_LABELING\_GUIDELINES.md` (v8) . `docs/HOW\_RALLY\_DETECTION\_WORKS.md` (2-layer mental model).

62 - ****Key finding (no re-investigation needed):**** the owner's "held-shuttle counted as a rally" bug is ****NOT** a missing capability** ? `backend/pipeline/detectors/rally\_gate.py` already

implements the net-crossing **"Gate A"** (`apply_gate(..., min_crossings=2)`) that kills service-feed/held-shuttle windows, but it is **only called from eval drivers** (and there was no `min_crossings` knob in `IndexingConfig`). **Wiring Gate A into the live segmenter** (now via `fusion_hybrid`) was the single cheapest, highest-confidence quality win. (P2 `FusionSegmenter` specifics: registry `fusion_hybrid`, **default-OFF**, seeds from substrate, persists `net_crossings` + optional person features, optional served Gate A/B via config knobs; adversarially reviewed; suite green. See `tests/test_served_gate_a_regression.py` for the honest finding that on production windowing Gate A is neutral/negative in some cases, hence kept opt-in with knob for safety, and the leaderboard revert note.)

63 - **Current status (2026-06-14):** P1 (person cue extraction to reusable `PersonDetector`) + P2 (`FusionSegmenter` + Gate A signal + served handoff gating) + experiment harness P1 DONE (adversarial review + NO-REGRESSION; suite green). **P3 next** (this PR + follow-ups): learned fusion (person features ? harness `compare` LOVO lift on the 6 golden videos; eager-person-compute optimisation); then P4 audio via `hit_detector`.

64 - **Pick-up order** (each = ONE PR off `master`, owner approval first, flag-gated default-OFF, promote only on measured LOVO):

65 1. (P2 first-step, largely done via `fusion_hybrid`) Wire Gate A into production served path + config knob (`min_crossings`); apply in hybrids at runtime. (Now available opt-in; served neutral on production windowing per litmus ? kept opt-in with knob for safety.)

66 2. P3 first step (this work item): harness `compare` LOVO litmus for person-on variant (`players_in_court` + `two_sided` + `colocation` etc. from `fusion_features`) on the 6 golden; add the eager compute optimisation note/landing. Pure + harness (testable here).

67 3. P3/P4 continuation: audio cue integration + full learned promotion if lift proven.

68 - **Discipline** (from `EXPERIMENT_HARNESS.md`): every new cue **flag-gated**, default OFF; promote to default **only on measured LOVO lift** through `run_regression.py` (exit 0/1/3); pinned `CONTROL` ? rollback = config flip. Harness re-uses ~80% existing code.

69 - ? Gate A, person cue logic, harness, and pure fusion paths are testable here; the heavier real-video / golden LOVO / ablation confirmation waits on the GPU-extracted golden features + `.rallies.csv` labels, which are not yet present in this checkout.

70 - **External due-diligence review inputs** (2026-06-17) ? PICK UP AFTER current ideas are exhausted; full details + citations in `RALLY_QUALITY_RESEARCH.md` §5.6. **An owner-commissioned 31-reference review of the quality report** *validated* the methodology and surfaced these **future** levers (research inputs, not active work, not new claims; verify citations before any submission):

71 1. **"First-mile" paper framing** (highest-value): stroke-classifiers (BST/DSTA) + the big datasets (ShuttleSet ~36k strokes, Fine-Badminton ~27k actions) all **assume a pre-trimmed rally clip already exists** ? we solve the ignored prerequisite. Pitch the paper as **"the missing untrimmed-video preprocessing pipeline that lets BST/MonoTrack run on raw footage."**

72 2. **Ground R2 in Action-Spotting / Precise-Event-Spotting** (SoccerNet/TennisSet/T-DEED) ? turns R2 into a defensible publishable contribution.

73 3. **Add domain baselines for any submission:** MonoTrack + BST on ShuttleSet / Fine-Badminton (we have only heuristic + Gemini today).

74 4. **Prefer TemporalMaxer over ActionFormer** for the M0.4 scorer swap (parameter-free max-pool, ~2.8x fewer GMACs, small-N robust) ? see "M0.4 next increment" below.

75 5. **Court polygons ? pose ? the rally-END lever:** masking on-court players can resolve *"occluded end-of-rally states"* (the #1 remaining gap vs Gemini), not just spectator noise.

76 6. **Name the eval/serve lesson "pipeline distribution skew"** ? a citable paper "lesson."

77
78 **Discipline reminder:** owner approval before starting any owner-gated item; one PR per step; babysit the PR (poll/fix/reply) until merge observed, *then* advance.

79
80 **## Prioritized next steps**
81 1. **[owner, in progress] GROW THE GOLDEN CORPUS** ? the single highest-leverage move; every lever below feeds on it.
82 Priority order for new videos: **(a) multi-court badminton** (the target distribution AND where the ship target
83 lives), **(b) ?3 matches per new sport ? table tennis first** (WASB transfers at F1 0.909; pickleball labels are
84 corpus-gold but not trainable until a clean MIT/Apache ball detector exists), (c) capture variety per
85 `VIDEO_RECORDING_GUIDELINES.md`; **adult sessions only for the training pool** (consent guardrail). Per video:
86 label ? `curate import-local --normalize --license Proprietary --fps <true fps>` ?

re-run the Gen-0 harness.

87 The paired sign test needs n: at n=10 with 9 wins, p?0.011 ? significance is reachable this month.

88 2. ****[owner decision] Set the ship-gate target**** ? now informed: floor = heuristic 0.480 (necessary), ****multi-court**

89 reference = wrapper 0.66** (the number that makes the owned model worth serving), clean-footage ceiling = 0.93

90 (not the target ? Gemini serves that tier). Suggested shape: "LOVO F1@tIoU0.5 ? 0.70 on multi-court folds with

91 paired-sign p<0.05 vs heuristic" ? owner to ratify/adjust.

92 3. ****M0.4 next increment (\$0/local)**** swap the Gen-0 LogReg for a lightweight temporal head (MIT, minutes on the

93 2070S ? NOT the heavyweight Gen-2 stack) inside the existing harness; per-video threshold calibration (the

94 LOVO-vs-oracle gap is visible per fold in the harness output). ****Prefer TemporalMaxer**** (parameter-free

95 max-pool, ~2.8x fewer GMACs, small-N robust) over ActionFormer/ASFormer per the 2026-06-17 external review

96 (`RALLY\_QUALITY\_RESEARCH.md` \$5.6) ? better fit for the cheap/local/fast mandate.

97 4. ****Multisport thread**** ingest Roora pickleball/TT CSVs (`import-local --normalize`); merged-prototype LOVO across

98 badminton+TT once TT has ?3 matches; identify a clean pickleball ball detector. ? TT-on-WASB serving stays gated

99 by C3 (provenance multiplies per sport).

100 5. ****PMF validation (cheaper than any engineering month)**** C13 academy interviews (+ the two added questions:

101 "multi-court footage nobody watches?" / "pay per-match to index it?"); camera pilot at the warmest 2-3 academies

102 (consent at capture; demo served via the Gemini path or human-polish ? the reel pattern exists:

103 `output/MahadevpuraSingles\_highlights.mp4`). ****Field kit COMPLETE (2026-06-12, owner ratified the**

104 physical-visit approach; reframed product-feedback-first per owner spec):**

105 `docs/PRODUCT\_FIELD\_ASSOCIATE\_FLYER.md` (the recruiting flyer ? owner fills ?/contact placeholders and recruits) +

106 `docs/FIELD\_VISIT\_PLAYBOOK.md` (the hired associate's visit playbook: coach + student script, demo-after-Q6

107 rule, ground rules, Phase-2 record-and-show-back stretch goal) +

108 `docs/FIELD\_VISIT\_ANSWER\_SHEET.md` (per-visit capture form incl. demo-reaction + student blocks) + `docs/PMF\_FIELD\_SIGNALS.md` (G/A/R anchors + ****pre-registered**

109 decision rules** ? PROCEED / PIVOT A-C / FALSIFIED thresholds fixed before the first visit).

110 6. ****Housekeeping**** ~~rotate the Gemini API key~~ ****DONE 2026-06-10**** (owner-verified working; confirm old-key

111 deletion in AI Studio); ****rename repo off "badminton"**** (owner action,

112 in BACKLOG); collector=SoR confirmation; off-peak niceties: none pending ? the battery completed.

113

114 **## Pointers / artifacts**

115 - Roadmap: `docs/OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md` . Topology: ADR-008 (`docs/archives/decisions/DECISIONS.md`) .

116 Study: `docs/archives/research/OWNED\_MODEL\_TRAINING\_STUDY.md` . Quality history:

117 `docs/archives/quality-iterations/2026-06-multivideo-lovo/`.

118 - Harness: `training/gen0/harness.py` (one command ? train ? LOVO ? gate ? artifact bundle; weights gitignored,

119 manifest reviewable). Featurizer: `backend/features/rally\_seq.py` (FEATURE_SCHEMA v1).

120 - Corpus: collector DB (system-of-record, 6 matches/262 rallies) . eval manifest `output/human\_lovo\_manifest.json`

121 (gitignored) . durable trajectories + Roora files: `C:\Users\avidu\Projects\Annotation Setup\Collect\Trajectories\` .

122 labels: `...\Collect\Golden Labelled\`.

123 - Gemini ops learnings: serial uploads (`indexing.ai\_max\_workers=1`) + chunk re-encode

124 (`indexing.ai\_chunk\_scale\_width=1280`) + 6-attempt exponential generate retry ? a cold/prepaid project burst-throttles

125 with a MISLEADING "credits depleted" 429; ramp gradually.
126 - ? No owned-model implementation beyond greenlit M0.x until owner greenlights the next
milestone. Committed next
127 PLATFORM slice = async job model (single-tenant).
128

7. user (2026-06-19T17:18:25.557Z)

1 - ROORA_LABELING_GUIDELINES.md ? Latest (v8) vendor
labeling guidelines mirrored from the collector's `docs/annotation/`, with the fusion-driven
court-calibration / singles-doubles additions marked `? ADDED (fusion)`.

2 - RALLY_QUALITY_RESEARCH.md ? **Foundation doc** for
closing the local-vs-Gemini rally-detection gap: the measured gap + the code-grounded over-merge
diagnosis ("shuttle moving ? rally"), an inventory of the eval/experiment harness + its gaps,
the strengthened evaluation framework (segmental Edit/F1@k that make over-segmentation first-
class, golden-corpus + active learning, Gemini-agreement as a *shadow* signal), an
adversarially-verified, cited external-research synthesis (TAL/TAS boundary heads, racket-
sports rally-state cues, small-data, eval rigor), a **prioritized cheap?durable roadmap** with
falsifiable success criteria, and (§7) a **tracked project plan** ? work-item checklist,
sequencing, the single quality **number** attributed to the effort, and a **definition of done**
(project completes when all tiers land + the number is recorded). The plan that the
QUALITY_ITERATIONS.md log executes against; ties together
EXPERIMENT_HARNESS / MULTI_SIGNAL_FUSION_PLAN / ANALYZERS_AND_RECONCILERS /
OWNED_MODEL_IMPLEMENTATION_PLAN.

3 - RALLY_DETECTION_GAP_CLOSURE.md ? **Tracked execution**
doc (2026-06-18) for closing the remaining rally-detection accuracy gap **per video**, before
launch, driven by the growing golden corpus. First ground-truth clip (GX010141) reframed it:
boundaries are tight (R2/R3/R4 holds) ? the gap is **detection**: (G1) recall (local & Gemini
both miss ~half on hard clips, and *different* rallies ? complementary), (G2) precision (local
over-fires), (G3) **rally-END over-extension** on the quick-return-after-point (shuttle still
moving ? needs a *player-state* signal). Levers (owner-gated): player-kinematics + shuttle-in-
hand / sustained-invisibility end-rule with reverse-timeline backtrack, `rally\_gate` Gate A
wiring, complementary fusion. **Peer** of `RALLY\_QUALITY\_RESEARCH.md` (execution tracker, not
research); §7 is the source of truth.

4 - PROMINENT_COURT_DETECTION.md ? **Design + experiment**
(2026-06-19, default-OFF, owner-gated flip) for the **multicourt G2 over-fire**: identify the
prominent (foreground) court (the one covering the majority of the frame) and keep only
rallies whose shuttle lives inside it ? re-defining the shuttle signal `(x,y,visible)` ?
`(x,y,visible-AND-inside-prominent-court)` **before** windowing, so a background/adjacent-court
shuttle never forms a rally. 2D floor polygon (MVP) ? **pseudo-3D play-volume** (refinement, so
high clears aren't clipped). Born from the GX010142 rally-#1 background-court FP; reuses
`fusion\_features.point\_in\_polygon` + `FusionConfig.court\_polygon`. §8 progress tracker is the
source of truth.

5 - RALLY_DETECTION_QUALITY_REPORT.md ? **Technical-**
paper checkpoint (2026-06-17) of rally-detection quality: the honest measured state (over-
segmentation milestone tolerance-F1 0.09170.323 at n=11, p=0.002; merge-rate 0.69470.150; first
real-footage ground-truth at-par with START at Gemini parity), the task-faithful R2 metric, a
detailed Gemini comparison, next steps + expected wins, areas yet to be explored, and a
publishability / path-to-submission verdict. Renders to a polished PDF; the externally-shareable
"toward a paper" summary of `RALLY\_QUALITY\_RESEARCH.md` + `QUALITY\_ITERATIONS.md`.

6 - EVALUATION.md ? How we measure and improve quality.

7 - EXPERIMENT_HARNESS.md ? Design (P1 SHIPPED) for A/B + shadow
+ SxS experimentation on rally-detection variants with **zero production regression** ?
experiments are decoupled from deploy (new cues default OFF, promote only on measured LOVO lift,
rollback = flip a flag). Composes the existing `backend/eval/` machinery (`calibration`,
`regression`, `metrics`); implemented in `backend/eval/experiment.py`.

8 - QUALITY_ITERATIONS.md ? **Living log** of rally-detection
quality iterations (hypothesis ? what shipped ? measured/expected effect ? lesson). The reverse-
chronological record; terminal-state snapshots get archived under `archives/quality-
iterations/`.

9 - ANALYZERS_AND_RECONCILERS.md ? Design (owner-gated,
NOT started): a **signal-fusion framework** for rally detection ? producers emit confidence-
scored **signals** at three temporal **scopes** (frame detectors / window analyzers / session
analyzers), a learned **scorer** weights them, and an auditable **report-first reconciliation**
pass lints the decision against the evidence. The spine: **no special-casing** in the decision

path** (signals?scorer, never `if/else`). Worked examples: a service-fault analyzer, a warm-up/practice span detector, and consuming a per-frame shuttle-state detector. A peer to `EXPERIMENT\_HARNESS.md` and `MULTI\_SIGNAL\_FUSION\_PLAN.md` (ADR-007 modular rally layer).

10 - PERSONALIZATION_PLAN.md ? Design (owner-adopted **hybrid**;
Phase-0 landing) for the **per-user personalization feature on a generalization-first
baseline**: a thin per-user *tuning* layer gated by the honest small-N statistics (a **min-N
promotion floor** + a **structural-guard exemption**, never a service gate) + a **contribution
flywheel** (free-tier videos pool into the owner-trained baseline; *a tool that gets better the
more you help it*). Records the **locked owner decisions** (identity, storage, `min\_n`
semantics, Gate-A default, free-tier pooling). Built so far: per-user promotion gate (#141) +
held-out-USER learning curve (#142).

11

12 - GOLDEN_REGRESSION_FIXTURES.md ? **Design (owner-
gated, NOT started)**: video-free, non-attributable regression fixtures so a clean clone runs
the rally-seg suite with **no video/GPU/DB**. Two layers ? the post-detector freeze + **dual
gates** (characterization + quality-floor) reusing the `backend/eval/` stack, and the **SEALED-
FIXTURES** privacy filter (de-attribution-at-source, tiered public/key-gated, shuttle-only
public tier). Carries a cited **IN/US/EU legal memo** (derived-data IP/privacy), a threat model,
and a **\$7 progress tracker + definition-of-done**. Extends
GOLDEN_DATA_SHARING.md.

13

14 ## Product / platform plans

15 - DESKTOP_APP_PLAN.md ? **Tracked scoping doc (2026-06-18, owner-
gated, `[design]` ? NOTHING built)**: package the local pipeline as a **cross-platform desktop
app** for non-technical players ? plug in the camera's USB/SD card, click once, get back **only
the highlight reels**, fully on-device (no upload, no original-video bloat). The centerpiece is
the **de-WSL native-compute port** (run WASB natively per-OS: Linux/Windows CUDA · macOS MPS/CPU
? drop WSL2), and its **make-or-break unknown**: is CPU acceptable on a typical *no-GPU*
enthusiast laptop (WASB is ~3.4× real-time on a laptop 2070S)? \$7 tracker = **P0 compute-
feasibility spike ? P1 native detector ? P2 app shell ? P3 packaging/installers**. Realizes the
L0 fully-local tier of VIDEO_LOCALITY_MODEL.md; is the
`LocalDesktop` ComputeBackend of PLATFORM_ARCHITECTURE.md §3.2;
the smaller/export-clean owned model
(OWNED_MODEL_IMPLEMENTATION_PLAN.md) is one answer to the
compute crux.

16

17 ## Practical / ongoing

18 - TRACKNET_WSL_SETUP.md

19 - Golden set docs (`GOLDEN\_SET`)

20 - Evaluation harness and calibration drivers (in `backend/eval/`)

21 - GOLDEN_DATA_SHARING.md ? How the GPU box and CPU dev/agent
sessions share golden artifacts (trajectory CSVs, `\*\_golden\_features.json`) via a OneDrive
folder; the GPU-extract ? CPU-litmus split. See **#175**.

22

23 ## Conventions & templates

24 - PROJECT_DOC_TEMPLATE.md ? Reusable skeleton for a **tracked
project doc**: status header · decisions-locked · progress tracker (??/??) · definition-of-done
· archived on completion. Standardizes the emergent pattern in `RALLY\_QUALITY\_RESEARCH.md` §7
and `GOLDEN\_REGRESSION\_FIXTURES.md`. Use for any substantial design/project work.

25

26 ## Historical documents

27 Completed plans, old research, ADRs, and checkpoints live in archives/.

28

29 - CODE_AUDIT_AND_TEST_HARDENING.md ? **COMPLETED**
(2026-06-14; T1?T5 + full loop + verification per #162). Living record of the June 2026 plumbing
/ contract / architectural audit, 626-test infrastructure, positive test plans for the 5
identified issues, multi-layer regression protocol (including ablation/experiment layer), and
pre/post-change verification checklist. See `docs/archives/CODE_AUDIT_AND_TEST_HARDENING-
COMPLETED-2026-06-14.md` and `docs/archives/checkpoints/2026-06-14-hardening-loop-complete.md`
for the final archived state. (Top-level is now pointer stub per review; full content in
archives/.)

30

31 See `archives/README.md` for the umbrella policy and `archives/past\_projects/README.md`
for classification of small tech debt / starter projects.

8. assistant (2026-06-19T17:18:28.547Z)

The README references most of my batch docs. Now let me read the 7 docs in my batch and the CLAUDE.md references.

9. user (2026-06-19T17:18:29.500Z)

```
1      # Platform Architecture ? Multi-User "Assistant Coach"
2
3      **Status:** ? Strategy / research ? for review. No code yet.
4      **Date:** 2026-06-07 . **Owner:** avidullu
5
6      > ? **Superseded where it disagrees with
[`OWNED_MODEL_IMPLEMENTATION_PLAN.md`](OWNED_MODEL_IMPLEMENTATION_PLAN.md)
7      > (2026-06-09, authoritative).** Specifically: (1) licensing is **MIT/Apache-2.0/BSD**
(not "Apache-2.0 only");
8      > (2) base model is **Qwen3-VL (Apache)** primary / InternVL3 (MIT) fallback ? **Gemma-4
is AMBER/bench-only**, not a
9      > cleared base; (3) the owned VLM is **not** the per-turn "conversational brain" ?
extraction (TAL head ? later VLM)
10     > feeds a per-video **event store + text-to-SQL** query layer ("extract once, query
many"); the VLM-answers-each-turn
11     > design is rejected. The $5A model/base/QA-harness detail below predates that and is
kept for context only.
12
13     **Purpose.** Map the path from today's **single-user, local** tool to a **multi-tenant
14     "assistant coach" platform**: users upload videos, get highlights **stored and
15     browsable by history**, and choose **where their data lives** (Google Drive / cloud
16     bucket / local) and **where the compute runs** (our GPU / their VM). The organizing
17     principle is to **decouple the layers** ? identity, ingestion, storage, orchestration,
18     execution/compute, the detection+highlight pipeline, and data-serving ? behind small
19     **pluggable interfaces**, reusing the registry pattern the codebase already uses for
20     segmenters/providers/validators/sports/annotations.
21
22     **Product framing (the north star *and* the v1 line).** This is an **assistant coach**,
23     not just a clipper ? but the strategy is **phased and unambiguous about scope**:
24     - **v1 (concrete, near-term):** **rally detection ? highlight-reel compilation ? per-
user
25     history ? a coach/user correction loop that feeds the golden set** ("assistant-coach
26     *lite"). Essentially today's pipeline made multi-user ? this is the real v1
deliverable.
27     - **Mid-term:** **rich rally/shot semantic metadata** (shot type, rally outcome) and the
28     **owned video-QA model** ($5A; $6 P4?P5).
29     - **North-star vision (strictly future ? not* v1):** **gait / biomechanics, per-athlete
30     identifying templates, and deep coaching feedback under a supervising coach** ($6 P6).
31     Inspiring, but explicitly deferred ? and the part that raises the **biometric**
privacy
32     bar ($7).
33
34     The buyer is the **academy/coach** (beachhead in
35     [`PMF_MARKET_RESEARCH.md`](PMF_MARKET_RESEARCH.md)); **power users self-serve**. Two
36     consequences ripple through the design: (1) the data model keeps an **optional coach ?
37     athlete overlay over a generic core** ($4.1) ? not baked in; (2) gait/biometrics is
38     **special-category data**, gated and deferred ($7).
39
40     > **Competitive note (validated by research):** none of Veo / Hudl / SwingVision /
41     > Spiideo offer true BYO-storage or BYO-compute ? they're vendor-cloud SaaS
42     > (SwingVision uniquely runs *edge* compute on the user's iPhone). So **"your video,
43     > your storage, your compute"** is a genuine differentiator for privacy-sensitive
44     > academies ? and SwingVision proves user-side compute is commercially viable. All four
45     > share the same **teams ? athletes ? clips** data model, which we adopt ($4).
46
47     > Scope: this is a **direction-setting doc**. It does not start implementation; the
48     > first infra slice (an async job model) is the on-hold #16 in
49     > [`DEFERRED_HARDENING_PLAN.md`](DEFERRED_HARDENING_PLAN.md). Commercial constraints
```

```

come
50     > from [ `COMMERCIALIZATION.md` ](COMMERCIALIZATION.md) (C4 paid-LLM no-train; C12 SaaS
51     > foundations) and `archives/MONETIZATION_AUDIT.md`.
52
53     ## 1. Current state & gaps (what we're evolving from)
54
55     | Concern | Today | Gap for multi-user |
56     |---|---|---|
57     | Identity / tenancy | none | no users, orgs, auth, or isolation |
58     | Storage | local `filepath` string; output to `output_dir` | no abstraction; can't BYO
Drive/bucket; raw video must be on the box |
59     | Data model | `videos` keyed by file **MD5**, `play_segments`, `events`,
`candidate_segments` (SQLite) | no owner/athlete link; **highlights aren't a first-class
persisted entity** (compile writes a file and returns a path); no per-user history |
60     | Execution | **async job model shipped** ? `/api/process` returns **202 + job_id**, a
single-worker `ThreadPoolExecutor` drains the `jobs` table, `GET /api/jobs/{id}` polls status |
single-tenant + local-only; can't dispatch to a user's VM/GPU yet |

```

10. user (2026-06-19T17:18:30.436Z)

```

1     # Hosting Plan ? khelsutra.guru
2
3     **Status:** PROPOSAL ? awaiting owner ratification.
4     **Date:** 2026-06-10 . **Constraints:** budget $10?25/mo . GCP credits available .
DigitalOcean
5     acceptable . alpha backend = owner's GPU box behind a tunnel . 10?25 alpha testers .
registrar = GoDaddy.
6     **Companion:** `docs/UI_PROPOSAL.md`.
7
8     ## 1. Recommendation in one line
9
10    **Alpha: free edge + your box** (Cloudflare DNS/Pages/Access/Tunnel ? FastAPI on the GPU
box) at
11    **$0/mo**, saving GCP credits; **Beta: GCP control plane + your box as pull-worker**
(Cloud Run +
12    GCS + Firebase Auth, credits-covered), keeping the edge where it's free.
13
14    Rationale for the Cloudflare pieces despite the GCP preference: GCP has no free
equivalent of the
15    Tunnel + Access combo (IAP requires a load balancer, ~$18/mo standing cost), and the
alpha needs
16    exactly three things GCP charges for but Cloudflare gives free: a secure tunnel to a
home box with
17    a custom domain, email-allowlist auth (free ?50 users ? fits 10?25 testers), and static
hosting.
18    Credits are better spent at beta on compute/storage, which is where GCP is excellent.
19
20    ## 2. Alpha topology ($0/mo)
21
22    ...
23    tester browser
24    ? https
25    ?
26    Cloudflare edge (free): DNS + TLS + Access (email-OTP allowlist)
27    ??? app.khelsutra.guru ? Cloudflare Pages (React SPA; auto-deploy from GitHub)
28    ??? api.khelsutra.guru ? Cloudflare Tunnel (outbound-only `cloudflared` service on
the box)
29
30    ?
31    ?
32    FastAPI :8000 on the GPU box (Windows)
33    ? Gemini segmenter path (no GPU in request path)
34    ? async jobs (#16), SQLite WAL, output/ on disk
35
36    - **No inbound ports opened** on the home network; `cloudflared` dials out. Home IP

```

```

never exposed.
37     - **Auth:** Cloudflare Access in front of BOTH hostnames; backend trusts the
38       `Cf-Access-Authenticated-User-Email` header (verify the Access JWT for rigor) for per-
user scoping.
39     - **Upload limit:** free-tier proxy caps request bodies at **100 MB** ? SPA uploads in
?90 MB
40       chunks (UI_PROPOSAL B2). Match videos (1?4 GB) flow fine as chunks.
41     - **Box offline = service offline.** Acceptable for alpha; SPA shows a friendly
"processing engine
42       offline" banner (a `/api/health` ping). Beta removes this with the queue.
43     - **Costs:** Pages, Tunnel, Access, DNS = $0. Only the Gemini per-run cost
(~$0.05/match) and
44       domain renewal.
45
46     ## 3. Beta topology (GCP credits; ~$15?20/mo post-credits)
47
48     ```
49     browser ??? Cloudflare edge (free, unchanged) ??? app.khelsutra.guru (Pages)
50                                     ??? api.khelsutra.guru ??? Cloud Run
(FastAPI control plane, scale-to-zero)
51                                     ? jobs
table (Cloud SQL Postgres)
52                                     ? signed
resumable URLs
53                                     ?
54                                     GCS buckets
(uploads / reels, lifecycle ? cold)
55                                     ? pull
jobs, outbound-only
56                                     GPU box = first
`byo_worker` (PLATFORM P3)
57     ```
58
59     - Uploads go **browser ? GCS directly** (resumable signed URLs) ? no proxy body limit,
no home
60     bandwidth for ingest; the box pulls only what it processes (WASB/owned-model jobs),
Cloud Run

```

11. user (2026-06-19T17:18:30.911Z)

```

1     # UI Proposal ? khelsutra.guru web app (alpha ? beta)
2
3     **Status:** PROPOSAL ? awaiting owner ratification (no implementation starts without
explicit approval, per CLAUDE.md).
4     **Date:** 2026-06-10 . **Inputs:** PMF_MARKET_RESEARCH, COMMERCIALIZATION,
PLATFORM_ARCHITECTURE, DEFERRED_HARDENING_PLAN (#16), ROADMAP, VIDEO_RECORDING_GUIDELINES,
reporting/spec.yaml, current API + static UI.
5     **Companion:** `docs/HOSTING_PLAN.md` (domain, topology, costs).
6
7     ## 1. Strategic frame ? why a UI now, and what it secretly is
8
9     The UI is not just product surface; it is the **corpus-growth engine**. The single
highest-leverage
10    item in NEXT_STEPS is growing the golden corpus, and the v1 scope already commits to a
11    **human correction loop**. Every tester who accepts/rejects/nudges a rally boundary in
the review
12    screen produces exactly the labels the Gen-0/M0.x track is starved for (subject to the
per-user
13    training opt-in guardrail). 10?25 alpha testers ? a steady stream of labeled, real-
world,
14    multi-court-biased footage ? the distribution where the owned model's 0.66 ship target
lives.
15
16    Serving economics per the 2026-06-10 quality table: alpha serves the **Gemini path**
(0.93 on clean footage, ~$0.05/run) ? no GPU required in the request path. The

```

WASB/owned-model

```
18     path stays on the owner's box and is *not* exposed in the alpha UI (segmenter choice
hidden).
19
20     **The CLI is not replaced.** The web app talks to the same FastAPI surface the CLI
drives; the CLI
21     remains the power-user/ops tool.
22
23     ## 2. Personas (from PMF research)
24
25     - **Coach / academy (primary, B2B, India):** batches of match recordings, wants rally
review,
26     per-player reels for students/parents, per-rally stats. Already records on static
cameras.
27     - **Enthusiast (B2C funnel):** records own matches, wants a highlight reel + basic
stats, shares
28     with friends. Tripod discipline is the known adoption risk ? the UI must teach
recording
29     guidelines *before* upload, not fail videos after.
30
31     ## 3. ALPHA ? "quick & dirty" (goal: 10?25 invited testers, end-to-end, zero hand-
holding)
32
33     **Stack:** React + Vite + Tailwind SPA, deployed per HOSTING_PLAN (static hosting, $0).
34     API = existing FastAPI on the GPU box via tunnel at `api.khelsutra.guru`.
35     **Auth:** Cloudflare Access email-allowlist (OTP), free ?50 users, **zero auth code** ?
backend
36     reads the authenticated-email header for per-user scoping (alpha-grade tenancy: header
is
37     trustworthy because the tunnel only accepts Access-authenticated traffic).
38
39     ### Alpha screens (6)
40
41     | # | Screen | What it does | Backed by |
42     |---|-----|-----|-----|
43     | A1 | **Matches** (home) | User's videos, status chip per video (verdict ?/??/?),
upload CTA | `videos` + owner scoping (B4) |
44     | A2 | **Upload & Process** | Drag-drop; inline recording-guidelines checklist (pre-
upload education from VIDEO_RECORDING_GUIDELINES); chunked upload; then auto-starts processing |
B2 upload endpoint ? B1 job |
45     | A3 | **Job progress** | Stage list (validation ? segmentation ? AI confirm ? report)
with live status; verdict banner on completion | B1 `GET /api/jobs/{id}` poll |
46     | A4 | **Rally review** ? | Video player + rally list; click rally ? seek & play that
span; per-rally **Accept / Reject / nudge start?end  $\pm 0.5$  s**; bulk "all good"; chips: duration,
shot count (+confidence) | B6 correction endpoints ? `candidate_segments.human_label/notes` ?
**the corpus engine** |
47     | A5 | **Highlights** | Select accepted rallies ? name reel ? compile ? download; list
of past reels | `/api/compile` + B3 download endpoint |
48     | A6 | **Report** | Renders the customer-audience report (verdict, notes, FAQ refs);
"nerd view" toggle shows technical.md | existing reporting artifacts |
49
50     ### Backend work the alpha actually requires (honest dependency list)
51
52     | ID | Item | Notes |
53     |---|-----|-----|
54     | B1 | **Async job model ? DEFERRED_HARDENING_PLAN #16** | The committed next PLATFORM
slice; `POST /api/process` ? 202 + `job_id`, `jobs` table (user_version 1?2),
ThreadPoolExecutor, `GET /api/jobs/{id}`. **Approving this proposal = greenlighting #16.** A
5?60 min blocking HTTP call cannot sit behind a tunnel UI. |
55     | B2 | **Chunked upload endpoint** | `POST /api/upload` accepting ?90 MB parts (free-
tier proxy body limit is 100 MB), assemble ? MD5 ? register under `allowed_video_root/<user>/`.
Size cap (e.g. 4 GB) + mp4/mov allowlist. |
56     | B3 | **Download endpoint** | `GET /api/highlights/{video_id}/{name}` streaming the
compiled reel (today `/api/compile` returns a server-local path the browser can't reach). |
57     | B4 | **Per-user scoping** | `owner_email` column on `videos` (migration), read from
```

the Access header, filter every query. Alpha-grade, single DB. |

58 | B5 | ****Retire/replace static UI**** | The SPA supersedes `backend/api/static` (which has a known `rallies` vs `play\_segments` field bug ? don't fix, replace). |

59 | B6 | ****Correction-loop endpoints**** | `PATCH /api/segments/{id}` ? accept/reject/boundary-nudge persisted to `candidate\_segments.human\_label`/`notes` (+ provenance: who, when). This is v1-scope "correction loop" made real. |

60 | B7 | ****Exposure hardening**** | CORS narrowed to the app origin; basic rate limit; upload quota per user; `allowed\_video\_root` set; Gemini key rotated (already in NEXT_STEPS housekeeping). |

12. assistant (2026-06-19T17:18:34.259Z)

These three are clearly LIVE design/proposal docs. Now let me read the remaining four and check the UI_ALPHA_EXECUTION_PLAN status (which NEXT_STEPS heavily references).

13. user (2026-06-19T17:18:35.197Z)

```

1      # UI Alpha Execution Plan ? KhelSutra Guru
2
3      **Status:** ACTIVE · updated 2026-06-10 · **Linked:** `UI_PROPOSAL.md` ·
`HOSTING_PLAN.md` ·
4      `mockups/ui_alpha_mockup.html` · ADR-009 (`archives/decisions/DECISIONS.md`).
5      **Sequencing principle (owner-requested):** build and test **everything locally first**;
6      deployment (DNS/tunnel/Pages/Access) starts only after the local end-to-end gate passes.
7
8      ## Decisions locked
9
10     | Date | Decision |
11     |---|---|
12     | 2026-06-10 | Product name **KhelSutra Guru** · domain **khelsutra.guru** (GoDaddy,
3-yr + protection) |
13     | 2026-06-10 | Separate **private repo `khelsutra`** ("heart and brain"): `web/` SPA
now, control plane at beta; backend enablers stay in the indexer repo |
14     | 2026-06-10 | **Local-first delivery** (this plan's phase structure) |
15     | open | Final "go" on PR-1 coding start (= #16 greenlight) · training opt-in default
OFF · invite list · stack (React+Vite+TS+Tailwind stands unless owner objects) |
16
17     ## Master checklist
18
19     ### G0 ? Setup (now)
20     - [x] Name/domain/repo-split ratified; spelling confirmed via GoDaddy receipt
21     - [x] `khelsutra` repo **scaffolded** at `C:\Users\avidu\Projects\khelsutra` (README,
.gitignore, web/, docs/)
22     - [x] **Owner:** git-init + private GitHub push ? done 2026-06-10:
23     `https://github.com/avidullu/khelsutra` (private; git runs Windows-side per ADR-009)
24     - [x] **Owner:** explicit "go" for PR-1 ? given 2026-06-10
25     - [x] **Owner:** Gemini API key rotated ? 2026-06-10, owner-verified (confirm old-key
deletion in AI Studio)
26
27     ### Phase L ? build & test locally (no cloud; ?6?7 sessions)
28
29     **L1 ? backend enablers** (indexer repo · branch off master · ONE PR each, no stacking ·
offline tests in every PR · GPU/real-video checks = owner):
30     - [x] **PR-1 · B1 async job model (#16):** `jobs` table (user_version 1?2), `POST
/api/process` ? 202 + `job_id`, `GET /api/jobs/{job_id}`, ThreadPoolExecutor worker,
restart/crash state handling ? **MERGED 2026-06-11, PR #87** (CI green: 333 passed; machine-side
real-video checks fold into L3)
31     - [x] **PR-2 · B2+B3 upload & download:** chunked upload (init ? sequential ?95 MB parts
? complete ? MD5; caps + mp4/mov allowlist; per-user dirs deferred to PR-3) + streaming `GET
/api/highlights/{video_id}/{name}` ? **MERGED 2026-06-11, PR #99** (also fixes static-UI
`segment_ids` compile bug + adds browser download link). **Live E2E verified at merge:** 86 MB
real video uploaded in 3 sequential parts and reassembled to MD5 byte-identity; out-of-order 409
/ traversal 400 (handler) + 404 (router) / unknown-id 404 confirmed; compile of a real confirmed
rally ? streamed ? download (3.9 MB reel).
32     - [ ] **PR-3 · B4+B7 identity & exposure:** `owner_email` column + scoping on every

```

```

query; Access-JWT/header read with `DEV_USER_EMAIL` env fallback for local dev; CORS narrowed;
rate limit; per-user quota
33   - [ ] **PR-4 · B6 correction loop:** `PATCH /api/segments/{id}` accept/reject/boundary-
nudge ? `candidate_segments.human_label`/`notes` + provenance (who/when); retire
`backend/api/static` (B5)
34
35   **L2 ? SPA** (khelsutra repo, `web/`; dev server proxies to `http://localhost:8000`):
36   - [ ] **M1 scaffold:** Vite + React + TS + Tailwind, router, typed API client, branding
(KhelSutra Guru)
37   - [ ] **M2:** A2 Upload (chunked, guidelines checklist) + A3 Job Progress (poll)
38   - [ ] **M3:** A1 Matches + A4 Rally Review (player seek, accept/reject/nudge wired to
PR-4)
39   - [ ] **M4:** A5 Highlights (compile + download) + A6 Report + engine-offline banner
40   - [ ] Unit tests for API client + review-state logic; `npm run build` clean
41
42   **L3 ? local end-to-end (owner's box, real match videos):**
43   - [ ] Full UI flow on localhost: upload ? job ? review/correct ? compile ? download
44   - [ ] `python run_all_tests.py` green; UI polling vs SQLite WAL sanity; WASB/CLI paths
untouched
45   - [ ] **GATE GL:** owner signs off local E2E ? only then deployment work begins
46
47   ### Phase D ? deployment (?2 sessions + ~1 hr owner)
48   - [ ] **D1 (owner):** Cloudflare account ? add khelsutra.guru ? GoDaddy nameserver
switch (HOSTING_PLAN $5) ? scoped API token ($6) ? Pages project connected to GitHub `khelsutra`
(build root `web/`)
49   - [ ] **D2 (Claude):** DNS records; `cloudflared` Windows service on the box ?
`api.khelsutra.guru`; Access apps + email-allowlist policies for `app` & `api`; SPA prod env
50   - [ ] **D3:** deployed smoke E2E (owner-only allowlist); go-live security checklist
(HOSTING_PLAN $7) all green
51   - [ ] **GATE GD:** checklist signed off
52
53   ### Phase R ? rollout
54   - [ ] Wave 1: 5 testers (C13/camera-pilot contacts) + feedback channel (WhatsApp group
or Google Form)
55   - [ ] 2-week retro ? punch list ? widen toward 25 ? beta planning (UI_PROPOSAL $4)
56   - Metrics: matches/week · % rallies corrected · median turnaround · verdict mix · week-2
tester retention
57
58   ## Working agreement
59   Claude (Cowork) designs, codes, writes tests + docs, and leaves a
`scratch/PRn_HANDOFF.md` per
60   work item. **Claude Code executes all git/PR mechanics** (branch, commit, push, PR,
conflict
61   fixes, merges after owner review) by reading that handoff ? the owner never copy-pastes
62   commands (ratified 2026-06-11 after PR-1/#88 friction; supersedes the command-block
pattern).
63   Cowork's sandbox cannot run git on mounts (ADR-009) and CI runs the offline suite on
every PR.
64   One PR per item; no stacking; never push to master directly. PowerShell note: no `&&`.
65
66   ## Risks & mitigations
67   Box uptime (offline banner, alpha-acceptable) · Gemini 429 burst-throttle (serial
workers,
68   `ai_max_workers=1`, 6-attempt backoff ? learned 2026-06-10) · CF free-tier 100 MB body
cap
69   (chunked upload design) · single-GPU box (alpha default = Gemini path; WASB jobs queue-
depth 1) ·
70   .guru renewal cost ?18.8k/3-yr (optional: transfer to Cloudflare Registrar at-cost after
the
71   60-day registration lock; domain-protection may need toggling for transfer).
72
73   ## Session map (default order)
74   S1?S2 PR-1 · S3 PR-2 · S4 PR-3 · S5 PR-4 · S6 M1+M2 · S7 M3 · S8 M4 · S9 L3 E2E + fixes
·
75   S10 D1?D3 · S11 wave-1 launch

```

14. user (2026-06-19T17:18:36.139Z)

```

1      # Storage-priced sharing ? mental model (DRAFT v2 for co-design)
2
3      **Status:** DRAFT ? owner + Claude co-design session pending; nothing here is ratified.
4      **Date:** 2026-06-12 (v2 adds: model-tier axis, social layer, on-device end-state, abuse
5      economics ? per owner's second brief) . **Source:** owner briefs (voice note `Umiya
Velociti.m4a`
6      pending transcription) + UI_PROPOSAL . HOSTING_PLAN . PMF research . NEXT_STEPS quality
table.
7      ? If ratified, this **supersedes the monetization framing in COMMERCIALIZATION.md**
(away from
8      per-video/subscription, toward storage-anchored tiers that bundle model quality) ?
record as an ADR.
9
10     ## 1. The one-line model (v2)
11
12     Don't sell videos or subscriptions ? **sell shelf space, and let model quality ride the
shelf**:
13     free = **2 GB + the owned model** ; paid = **more shelf + the best model** ; the visible
meter is
14     always storage. The social layer makes the shelf worth filling; the owned model makes
the free
15     tier cost ~nothing; the end-state moves compute onto the customer's own hardware
entirely.
16
17     ## 1a. The three coupled axes (one user-visible meter)
18
19     | Axis | Free | Paid | End-state (serious users) |
20     |---|---|---|---|
21     | **Storage** (the meter) | 2 GB library | 10 GB+ / pooled | Mostly local disk; cloud =
shareable library only |
22     | **Model** (rides the tier) | Owned model only ? **never Gemini** | Best path (two-
stage refine / frontier) | Best owned model, on-device |
23     | **Compute** (the cost) | Our GPU, capped runs | Our GPU, priority | **Their GPU**
(USB-drive workflow) |
24
25     ## 2. The load-bearing decision: two tiers of object
26
27     | Tier | What | Size reality | Stored? | Quota? |
28     |---|---|---|---|---|
29     | **Source** (raw match upload) | Processing INPUT | 1?4 GB each | Temporarily: kept
through processing + correction window, then **auto-deleted** (e.g. 7 days or on "corrections
confirmed") | Not by default; optional "keep my raw match" toggle = counts |
30     | **Library** (compiled reels, kept rally clips, thumbnails) | The PRODUCT | reel ?
50?80 MB (3?5 min 720p) | Durable | **Yes ? this is what 2 GB means** |
31
32     Why this split: 2 GB cannot hold even one raw match, but holds **~25?40 reels** ? the
quota is
33     only meaningful against outputs. It also kills the "free CDN" abuse vector for a
storage-priced
34     product: **the library only ever contains pipeline outputs** ? there is no path to park
arbitrary
35     files in it. (Sources additionally pass the existing relevance validator.)
36
37     So: **Do we store the uploaded video?** ? **Yes, transiently ? it's a processing input
with a
38     retention deadline, not a possession.** The reel is the possession.
39
40     ## 3. Eviction policies = a pluggable registry (the repo's own pattern)
41
42     `EvictionPolicy` ABC + registry, exactly like `segmenter_registry`:
43     `fifo` (oldest first) . `largest_first` . `smallest_first` . `least_visited` (view-

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```

counter) .
44     `hard_stop` (no deletion ? block new saves until user frees space or upgrades).
45
46     - **Proposed default: `hard_stop`** ? never silently delete someone's match memories;
FIFO etc.
47     are opt-in conveniences. (Co-design question #2.)
48     - **Grace flow:** over quota ? banner + email with the exact list the chosen policy
would evict ?
49     N-day grace (propose 7) ? evict, logged in an `eviction_log`. Offer download-before-
evict.
50     - `least_visited` needs a decision: owner views, public views, or both?
51
52     ## 4. CUJ after upload (end-to-end)
53
54     ```
55     record ? upload source ? async job (PR-1) ? review/correct (corpus engine, PR-4)
56         ? compile reel(s) ? reel lands in LIBRARY (quota meter ticks)
57         ? share: profile page / unlisted link / private
58         ? source hits retention deadline ? auto-delete
59         ? library overflow ? eviction policy (with grace)
60     ```

```

15. user (2026-06-19T17:18:36.656Z)

```

1     # Video locality ? where the bytes live (mental model + open questions)
2
3     **Status:** THINKING DOC (owner-requested, 2026-06-12) ? a shared mental model for
4     the upload/storage problem, with the open questions named. **No implementation is
5     greenlit by this doc** (per CLAUDE.md, both tracks stay owner-gated).
6     **Companions:** `UI_PROPOSAL.md` (B2/B3 upload/download), `HOSTING_PLAN.md`,
7     `PLATFORM_ARCHITECTURE.md` §3 (future StorageBackend/ComputeBackend registries),
8     collector ADR-002 (the frame-identical proxy contract this doc leans on).
9
10    ## 0. The questions that triggered this (and the short answers)
11
12    ***"Do we store the video they upload?"** Yes ? today the chunked-upload flow
13    assembles the full file under `security.upload_dir` (default `output/uploads`) on
14    our box, registers it by MD5, and nothing ever deletes it (retention policy is an
15    acknowledged gap ? BACKLOG). Processing also strews derived artifacts (chunks,
16    frames, trajectories, WSL temp) and, on the Gemini path, uploads ?1280px
17    re-encoded chunks to Google's Files API.
18
19    ***"How would they upload a 10 GB file?"** Today they can't: `/api/upload/init`
20    rejects anything over `security.max_upload_mb` (default **4096 MB**), and even
21    under the cap the file moves as sequential ?90 MB parts through a free-tier edge
22    whose body limit forced that part size. For scale: 10 GB ? 30?45 min of 4K phone
23    video (or ~2?3 h at 1080p); on a typical Indian uplink that's **~1 h at 20 Mbps,
24    ~4.5 h at 5 Mbps** ? before retries. Browser-tab uploads of that size are
25    miserable everywhere; expecting academies to manage S3/GCS buckets is a
26    non-starter. **The 10 GB problem is real and the current design doesn't solve it.**
27
28    ***"Can we make it so there is NO upload?"** Mostly ? because of §1.
29
30    ## 1. The central insight: timestamps are the product; the video is dead weight
31
32    Walk the pipeline and ask what each stage actually consumes:
33
34    - The **Gemini segmenter** never sees the original ? it re-encodes chunks down to
35    **1280 px** (`indexing.ai_chunk_scale_width`) before uploading them to Google.
36    - The **detectors/featurizer** consume downsampled frames and trajectories.
37    - The only consumer of the original full-res bytes is the **highlight compiler**,
38    and only to cut clips ? a pure-ffmpeg operation that can run **wherever the
39    original lives**.
40    - What the customer receives is `{rally intervals, report, reel}`. Intervals +
41    report are **kilobytes**. Given the intervals, the reel can be cut on the

```



```

42     machine that already has the original.
43
44     So: **any design that moves the 10 GB is moving the wrong thing.** The question
45     is never "how do we upload the video" but "what is the smallest artifact that
46     must cross the network for this tier of service."
47
48     ## 2. The locality ladder
49
50     From most to least bytes leaving the academy's Windows desktop. Sizes are rough
51     ratios for a 4K source ? measure on a real file before committing to any tier.
52
53     | | What crosses the network | ~Bytes (10 GB source) | What it needs | Status |
54     |---|---|---|---|---|
55     | **L3 ? full upload** (today's flow) | everything | 10 GB (blocked: 4 GB cap) | paid
edge tier + resumable uploads + server disk + retention policy | built, capped, wrong for big
files |
56     | **L2 ? proxy upload** | a timeline-identical ?1080p re-encode | ~1?2.5 GB | a tiny
client prep step (ffmpeg) ? **the exact frame-identical contract we already built and verified
as collector `prep-annotation-proxy`** (same fps, same frame count, parity-checked, MD5?MD5
manifest) | contract proven; client wrapper + server acceptance = new work |
57     | **L1 ? chunks/candidates upload** | only the ?1280px AI chunks (or just candidate
windows) the Gemini path needs anyway | ~0.3?1 GB | client-side chunk+downscale (ffmpeg, CPU);
server becomes a **GPU-less orchestrator** | the server half exists (it's the gemini segmenter's
own flow, relocated) |
58     | **L1-direct** | chunks go **straight from the client to Gemini** ? zero video bytes
through our server | ~0 through us | per-user key custody / metering / cost ceilings (OQ4) |
idea only |
59     | **L0 ? fully local** | nothing (calls to Google only if the Gemini path is used) | 0 |
the owned on-device model (roadmap north star) or a local GPU + our stack | not alpha-feasible;
the long-term answer |
60     | **L4 ? sneakernet** | a hard drive | 0 over network | a person (the field associate /
owner) | how the pilot actually works today |

```

16. assistant (2026-06-19T17:18:40.233Z)

All four are clearly LIVE (DRAFT/THINKING/ACTIVE design docs, owner-gated future work).
 UI_ALPHA_EXECUTION_PLAN is ACTIVE with PR-3/PR-4 unchecked. Now let me read COMMERCIALIZATION.md
 and check the CLAUDE.md references + inbound link counts for all 7.

17. user (2026-06-19T17:18:41.661Z)

```

1     # Commercialization Tracker
2
3     **Status:** ? LIVING ? track this in parallel with product/quality work.
4     **Owner:** avidullu . **Last refreshed:** 2026-06-07
5
6     **Purpose.** Single source of truth for the **commercial** dimension of the project:
7     licensing/legal posture, codec & output economics, hosting unit-economics, the
8     golden-set vendoring spend, packaging/distribution, and the product-market-fit
9     risks that gate monetization. Updated as decisions land ? see the
10    [Changelog](#changelog) at the bottom.
11
12    **Merged from / supersedes:**
13    - `archives/MONETIZATION_AUDIT.md` (2026-05-30 deep-research licensing & economics
14      audit) ? kept as the **research-of-record** (per-claim verification transcript,
15      primary sources). This doc carries its *current* status forward.
16    - `archives/COMMERCIAL_READINESS_REVIEW.md` (code-health + PMF assessment) ?
17      archived; its commercial findings + the quality-phase priorities live here now.
18
19    **Premise (unchanged, verified):** the product is a **closed-source hosted API** ?
20    users upload videos to a cloud NVIDIA GPU VM and get highlight reels back. We never
21    distribute code/binaries, so GPL/LGPL *distribution* copyleft is not triggered ? but
22    **AGPL §13 (network clause)**, **metered-API terms**, and **video-codec patent
23    royalties** all still apply.
24

```

```

25      ---
26
27      ## 1. Commercialization dashboard
28
29      Legend: ? done . ? in progress / watch . ? blocker / needs action . ?? needs legal
counsel . ? future
30      Priority: **P0** (do first) . **P1** . **P2** (lowest). Rows are grouped **Pending ? In-
flight ? Completed**, each sorted by priority. (C# IDs are stable identifiers ? they don't imply
order.)
31
32      | # | Workstream | Pri | Status | Where it stands | Next action / owner decision |
33      |---|---|---|---|---|---|
34      | | **? PENDING (needs action / not started)** | | | |
35      | C5 | **Output codec royalties** | **P0** | ? | Highlights are still encoded **H.264 +
AAC** (`libx264` in `backend/pipeline/stitcher/compiler.py`) ? encode-side AVC patent exposure
remains. | Switch output to **AV1 + Opus** (royalty-free) on an **Ada-class GPU (L4/L40)** for
HW encode. See §4. Decision: provision Ada-class GPU? |
36      | C9 | **Packaging / distribution** | **P0** | ? | `pyproject.toml` (PEP 621, hatchling)
added; the diagnostic `setup.py` renamed to `scripts/doctor.py`; `indexer` console entry point;
CI installs `pip install -e .[dev]`. torch/torchvision stay an out-of-band `--index-url` install
(PEP 621 can't pin CUDA wheels). | Implemented per `DEFERRED_HARDENING_PLAN.md` (#7). Remaining:
publish `obsreport` so the `reporting` extra is pip-installable. |
37      | C12 | **SaaS foundations (async/auth/tenancy)** | **P1** | ? | Async job model shipped
(`DEFERRED_HARDENING_PLAN.md` #16, PR #87); auth/tenancy still future. | Multi-user/platform
strategy in `PLATFORM_ARCHITECTURE.md` (pluggable storage/compute, tenancy, orchestration). |
38      | | **? IN-FLIGHT (in progress / watch)** | | | |
39      | C4 | **Runtime AI strategy (A/B/C + owned model)** | **P0** | ? | Live = **(C) Gemini
2.5 Flash, paid tier**. Long-term = **own a self-hosted generative video-QA model** (fine-tuned
open VLM) as primary, **paid Gemini always-ready as fallback** via the provider registry. | Make
"provider registry + paid fallback" a core principle; build the owned model **after** the
scaffolding/flywheel are solid. See `PLATFORM_ARCHITECTURE.md` §5A. |
40      | C13 | **Beachhead validation** | **P0** | ? | Market research
(`PMF_MARKET_RESEARCH.md`) recommends **India badminton academies** as the #1 beachhead.
Unvalidated. | Cheapest tests: 10?15 academy/coach discovery interviews; rally-detection P/R
benchmark on 100+ real amateur clips; fake-door pricing (B2C ~$99/yr, B2B academy ~$600/yr). See
§6. |
41      | C8 | **Golden-set vendoring spend** | **P1** | ? | ADR-006: **Roora (Bengaluru)**
selected for an initial paid pilot; licensed/owned video only (no user uploads to vendors). |
Confirm quality-per-dollar from the pilot; then GS-3 `HumanVendorProvider`. See
`GOLDEN_SET_VENDORING.md`. |
42      | C3 | **Shuttle-tracking weights provenance** | **P1** | ? ? | WASB-SBDT **code is MIT
(clear)**; distributed **checkpoints are academic-data-trained** ? not covered by the MIT code
grant. MonoTrack/YOLO-NAS banned (noncommercial). | **Ship-gating** blocker (multiplies per
sport). Resolve via an own-trained detector **or** a clean MIT/Apache detector swap ? NOTE:
ADR-002 says our temporal labels can't train the detector (needs coordinate labels), so own-
weights is a new, un-scoped workstream, NOT delivered by ROADMAP Phase 4. |
43      | C14 | **Owned-model base licensing + training-data policy** | **P1** | ? ? | Owned
video-QA model must build on a **commercial-clean (MIT/Apache-2.0/BSD), video-capable** base ?
primary **Qwen3-VL (Apache-2.0)**, fallback **InternVL3 (MIT)**. **Gemma 4 = AMBER / bench-
only** (Apache grant likely but Prohibited-Use posture unconfirmed ? counsel needed; not a build
target yet). **Reject Gemma 3** (viral derivative clause ? permanent Google PUP), Llama Vision
(naming + 700M-MAU), `gpt-oss` (text-only). **Hard rule: never train on paid-LLM
(Gemini/OpenAI/Anthropic) outputs** (competing-model clause). The moat is the **consented
dataset + eval harness**, not the model. | **MIT/Apache-2.0/BSD** for code+data+weights; **open-
model teachers only**; separate explicit **training opt-in**; **exclude minors**; per-clip
consent ledger. See `OWNED_MODEL_IMPLEMENTATION_PLAN.md` (authoritative). |
44      | C6 | **Inbound HEVC decode** | **P2** | ? ? | Decoding users' GoPro H.264/**HEVC**
uploads is unavoidable; HEVC pools are fragmented + highest-uncertainty. | Low/uncertain
residual; confirm decode-side exposure with counsel before launch. |
45      | C7 | **Hosting unit-economics** | **P2** | ? | Per-video cost is *cheap* for Gemini
(~$0.05 input for a 10-min clip); risk is volume × latency, not unit price. Self-host/offline
costs unmeasured. | **Benchmark $/video & latency on the target VM** for A/B/C before pricing.
See §5. |
46      | C10 | **Value layer (product surface)** | **P2** | ? | Minimal exports shipped
(EDL/JSON/CSV + `basic_stats`, PR #51). | Deepen: per-rally tags/notes that feed golden, per-

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player/match stats, clip gallery. See §6. |
47      | C11 | **Input-quality enforcement** | **P2** | ? | Per-video observability reports
surface a customer verdict + footgun flags incl. audio-present (PR #50). | Turn more
`VIDEO_RECORDING_GUIDELINES` dimensions into validators/report rules. |
48      | C2 | **`supervision` ByteTrack pin** | **P2** | ? | Pinned `<0.30.0` (deprecation
ahead). | Migrate before lifting the pin (BACKLOG). Not a legal risk, a maintenance one. |
49      | | **? COMPLETED** | | | |
50      | C1 | **AGPL / copyleft removal** | ? | ? | ADR-005 shipped: AGPL `ultralytics` YOLOv8
replaced with **torchvision Faster R-CNN (BSD) + `supervision` ByteTrack (MIT)**. No AGPL
code/weights in the stack. | Done. Keep new deps permissive. |
51
52      ---
53
54      ## 2. Licensing & dependency posture (refreshed risk table)
55

```

18. user (2026-06-19T17:18:45.094Z)

```

=== PLATFORM_ARCHITECTURE ===
(no archived copy at archives/ top level)
=== HOSTING_PLAN ===
(no archived copy at archives/ top level)
=== UI_PROPOSAL ===
(no archived copy at archives/ top level)
=== UI_ALPHA_EXECUTION_PLAN ===
(no archived copy at archives/ top level)
=== STORAGE_SHARING_MODEL ===
(no archived copy at archives/ top level)
=== VIDEO_LOCALITY_MODEL ===
(no archived copy at archives/ top level)
=== COMMERCIALIZATION ===
(no archived copy at archives/ top level)
=== nested archives ===
docs/archives/checkpoints:
2026-06-14-hardening-loop-complete.md
2026-06-eval-baseline
2026-06-logging-reporting
2026-06-scaffolding-complete
2026-06-strategy-foundation

docs/archives/past_projects:
README.md
low-regret-moves

docs/archives/research:
AUDIO_E1_FINDINGS.md
AUDIO_RALLY_DETECTION_PLAN.md
OWNED_MODEL_TRAINING_STUDY.md
RALLY_DETECTION_RESEARCH.md

```

19. assistant (2026-06-19T17:18:51.636Z)

No archived copies exist for any of the 7 docs (no DEDUPE candidates). COMMERCIALIZATION is explicitly LIVING and cited by CLAUDE.md as authoritative strategy. Now let me grep for inbound references across the whole repo and check CLAUDE.md citations.

20. user (2026-06-19T17:18:53.699Z)

```

CLAUDE.md:2
docs\archives\DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md:1
docs\archives\README.md:1
docs\archives\COMMERCIAL_READINESS_REVIEW.md:1
docs\archives\decisions\DECISIONS.md:2
docs\archives\MONETIZATION_AUDIT.md:1

```

docs\NEXT_STEPS.md:1
docs\archives\checkpoints\2026-06-strategy-foundation\README.md:2
docs\VIDEO_LOCALITY_MODEL.md:2
docs\DESKTOP_APP_PLAN.md:6
docs\UI_PROPOSAL.md:1
docs\EXECUTION_POLICY.md:1
docs\EXPERIMENT_HARNESS.md:1
docs\archives\past_projects\low-regret-moves\LOW_REGRET_MOVES_PLAN.md:1
docs\UI_ALPHA_EXECUTION_PLAN.md:1
docs\HOSTING_PLAN.md:1
docs\COMMERCIALIZATION.md:2
docs\I18N_PLAN.md:4
docs\README.md:1
docs\STORAGE_SHARING_MODEL.md:2
docs\PMF_MARKET_RESEARCH.md:1
docs\PLATFORM_ARCHITECTURE.md:2

Found 37 total occurrences across 22 files.

21. user (2026-06-19T17:18:55.236Z)

13: - **Strategy (post-scaffolding):** `docs/COMMERCIALIZATION.md`,
`docs/PMF\_MARKET\_RESEARCH.md`,
14: `docs/PLATFORM\_ARCHITECTURE.md`, `docs/DEFERRED\_HARDENING\_PLAN.md`. Also `docs/ROADMAP.md`

22. assistant (2026-06-19T17:18:58.262Z)

CLAUDE.md cites both COMMERCIALIZATION.md and PLATFORM_ARCHITECTURE.md as live strategy docs.
Let me get the per-file referrer breakdown to populate key_referrers and confirm citations in
DESKTOP_APP_PLAN, I18N_PLAN, EXECUTION_POLICY.

23. user (2026-06-19T17:18:59.342Z)

Found 9 files
docs\README.md
CLAUDE.md
docs\DESKTOP_APP_PLAN.md
docs\COMMERCIALIZATION.md
docs\EXECUTION_POLICY.md
docs\EXPERIMENT_HARNESS.md
docs\VIDEO_LOCALITY_MODEL.md
docs\I18N_PLAN.md
docs\archives\checkpoints\2026-06-strategy-foundation\README.md

24. user (2026-06-19T17:18:59.843Z)

Found 4 files
docs\VIDEO_LOCALITY_MODEL.md
docs\UI_ALPHA_EXECUTION_PLAN.md
docs\archives\decisions\DECISIONS.md
docs\UI_PROPOSAL.md

25. user (2026-06-19T17:19:00.751Z)

Found 3 files
docs\VIDEO_LOCALITY_MODEL.md
docs\UI_ALPHA_EXECUTION_PLAN.md
docs\HOSTING_PLAN.md

26. user (2026-06-19T17:19:01.245Z)

Found 4 files
docs\NEXT_STEPS.md
docs\archives\DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md

docs\I18N_PLAN.md
docs\archives\decisions\DECISIONS.md

27. user (2026-06-19T17:19:02.168Z)

No files found

28. user (2026-06-19T17:19:02.634Z)

Found 2 files
docs\README.md
docs\DESKTOP_APP_PLAN.md

29. user (2026-06-19T17:19:03.095Z)

Found 11 files
CLAUDE.md
docs\DESKTOP_APP_PLAN.md
docs\archives\README.md
docs\PLATFORM_ARCHITECTURE.md
docs\STORAGE_SHARING_MODEL.md
docs\I18N_PLAN.md
docs\archives\past_projects\low-regret-moves\LOW_REGRET_MOVES_PLAN.md
docs\archives\checkpoints\2026-06-strategy-foundation\README.md
docs\archives\COMMERCIAL_READINESS_REVIEW.md
docs\archives\MONETIZATION_AUDIT.md
docs\PMF_MARKET_RESEARCH.md

30. assistant (2026-06-19T17:19:09.859Z)

I have all the evidence I need. Notable findings:

- ****STORAGE_SHARING_MODEL.md**** has zero inbound references (it's a DRAFT not yet wired into README/NEXT_STEPS) ? but it's an active DRAFT co-design doc, so KEEP_LIVE (zero inbound = no link-breakage risk, not a staleness signal here).
- ****UI_ALPHA_EXECUTION_PLAN.md**** is ACTIVE with PR-3/PR-4 unchecked and the SPA phase (L2/L3/D/R) entirely unbuilt ? clearly in-progress.
- All 7 docs are LIVE: strategy/proposal/draft/thinking/active-execution. None terminal. No archived duplicates exist. COMMERCIALIZATION and PLATFORM_ARCHITECTURE are cited in CLAUDE.md as live strategy.

All 7 are KEEP_LIVE with no contradiction trap (none claim COMPLETED/DONE/REJECTED). Returning the structured result.

31. user (2026-06-19T17:19:41.769Z)

Structured output provided successfully

32. assistant (2026-06-19T17:19:52.752Z)

All 7 docs in my batch audited. Summary of verdicts (all KEEP_LIVE ? none are in a terminal state, none have archived duplicates):

- ****PLATFORM_ARCHITECTURE.md**** ? Strategy/research, "No code yet"; cited live in CLAUDE.md. Partially superseded-where-it-disagrees but explicitly kept for context. KEEP_LIVE.
- ****HOSTING_PLAN.md**** ? PROPOSAL awaiting ratification; companion to the active UI-alpha track. KEEP_LIVE.
- ****UI_PROPOSAL.md**** ? PROPOSAL awaiting ratification; defines the A1?A6 / B1?B7 scope the active execution plan runs against. KEEP_LIVE.
- ****UI_ALPHA_EXECUTION_PLAN.md**** ? Status ACTIVE; PR-3/PR-4 + the whole SPA/deploy/rollout phases unchecked; cited live in NEXT_STEPS. Clearly in-progress. KEEP_LIVE.
- ****STORAGE_SHARING_MODEL.md**** ? DRAFT, "nothing here is ratified," co-design pending. Zero inbound refs (no link-breakage risk) but it's un-wired future design, not terminal. KEEP_LIVE.
- ****VIDEO_LOCALITY_MODEL.md**** ? THINKING DOC, "No implementation is greenlit"; anchors the L0

tier of the active desktop-app track. KEEP_LIVE.

- ****COMMERCIALIZATION.md**** ? Self-declared ? LIVING strategy SoR, 11 inbound refs, cited first in CLAUDE.md's live strategy list. KEEP_LIVE.

No ARCHIVE_NOW, no ASK_OWNER (no doc claims COMPLETED/DONE/REJECTED, so no contradiction trap fired), and no DEDUPE_STUB (Glob of docs/archives confirmed none of these 7 stems have an archived copy). Relevant files all under `C:\Users\avidu\Projects\badminton-highlight-indexer\docs\`.